

TEST CASE DOCUMENT

Smart Student Productivity Platform

1. Document Control

Field	Details
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1.0	24/09/2026	Tech lead	Initial Test Case Document	Draft

3. Approval

Role	Name	Signature	Date
Project Manager / BA	Sakshi Shingole		
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4. Introduction

The Smart Student Productivity Platform is a centralized academic web application designed to manage academic activities and productivity for students, faculty members, HODs/domain administrators, and super administrators.

The platform provides academic management, assignments, examinations, attendance, learning resources, communication, online classes, AI-based doubt assistance, study groups, notifications, reporting, and administrative management.

The purpose of this Test Case Document is to define the test scenarios and test cases required to verify that the implemented system meets the approved functional and non-functional requirements.

5. Testing Objectives

The objectives of testing are:

1. Verify that all major system functionalities work as expected.
 2. Verify authentication and authorization.
 3. Verify role-based access control.
 4. Verify Student Portal functionality.
 5. Verify Faculty Portal functionality.
 6. Verify HOD/Domain Admin functionality.
 7. Verify Super Admin functionality.
 8. Verify academic data relationships.
 9. Verify assignment and assessment workflows.
 10. Verify attendance and result management.
 11. Verify AI Doubt Solver functionality.
 12. Verify study group functionality.
 13. Verify notifications and communication.
 14. Verify file upload and storage functionality.
 15. Verify security controls.
 16. Verify integration between frontend, backend, Firebase and external services.
 17. Verify error handling and validation.
 18. Verify responsive and usable interfaces.
 19. Verify critical business rules.
 20. Provide evidence for UAT and final acceptance.
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6. Scope of Testing

6.1 In Scope

The following modules are included:

- Login and authentication
- Role identification
- Authorization
- Student Portal
- Faculty Portal
- HOD Portal
- Super Admin Portal
- Academic structure
- Courses and subjects
- Assignments
- Assessments
- Projects
- Examinations
- Attendance
- Results
- Learning materials
- Tasks
- Calendar
- Communication
- Online classes
- Notifications
- AI Doubt Solver
- Faculty doubt escalation
- Study groups
- Exam-focused study
- User management
- Role and permission management
- Reports
- Audit logs
- File management
- Firebase integration
- Cloudinary integration
- Cloud Functions
- Security Rules
- FCM notifications
- Google Meet integration where implemented

6.2 Out of Scope

Any functionality not defined in the approved BRD/SRS is outside the scope of this test document.

8. User Roles

Role	Main Responsibility
Student	Access academic information and productivity features
Faculty	Manage assigned academic activities
HOD / Domain Admin	Manage department/domain academic activities
Super Admin	Manage platform-wide users, structure and configuration

11. Authentication Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-AUTH-001	FR-AUTH-001	Login with valid student credentials	Valid student email/password	1. Open login page. 2. Enter valid email. 3. Enter valid password. 4. Click Login.	User is authenticated successfully and redirected to the Student Portal/Dashboard.	Critical
TC-AUTH-002	FR-AUTH-001	Login with invalid password	Valid email + incorrect password	1. Enter valid email. 2. Enter incorrect password. 3. Click Login.	Login is rejected and an appropriate error message is displayed. User remains on login page.	High
TC-AUTH-003	FR-AUTH-001	Login with empty email	Email = blank	1. Leave email blank. 2. Enter password. 3. Click Login.	Email validation message is displayed. Authentication request is not submitted.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-AUTH-004	FR-AUTH-001	Login with empty password	Password = blank	1. Enter valid email. 2. Leave password blank. 3. Click Login.	Password validation message is displayed. Authentication request is not submitted.	High
TC-AUTH-005	FR-AUTH-001	Login with invalid email format	student@	1. Enter invalid email format. 2. Enter password. 3. Submit.	System displays email-format validation and does not submit invalid credentials.	Medium
TC-AUTH-006	FR-AUTH-002	Logout	Active session	1. Open portal. 2. Click Logout.	Firebase session is terminated and user is redirected to login page.	Critical
TC-AUTH-007	FR-AUTH-003	Direct access to protected student page without login	/student/dashboard.html	1. Open protected URL directly.	User is redirected to login/authentication page. Protected data is not displayed.	Critical
TC-AUTH-008	BR-12	Login using deactivated account	Deactivated user credentials	1. Enter credentials. 2. Click Login.	Authentication/access is rejected and user cannot enter the portal.	Critical
TC-AUTH-009	FR-AUTH-004	Password visibility toggle	Valid password	1. Enter password. 2. Click visibility icon.	Password visibility changes without altering the entered password.	Low
TC-AUTH-010	FR-AUTH-005	Session persistence after page refresh	Active session	1. Login. 2. Refresh page.	Authorized session remains valid according to configured authentication/session behavior.	High

2. Role-Based Access Control Test Cases

The platform has four defined roles: **Student, Faculty, HOD/Domain Admin, and Super Admin.**

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-RBAC-001	BR-03	Student accesses Student Portal	Student account	Navigate to student dashboard	Student dashboard loads successfully.	Critical
TC-RBAC-002	BR-03	Student attempts Faculty Portal access	Faculty URL	Enter Faculty Portal URL directly	Access is denied or user is redirected to authorized portal.	Critical
TC-RBAC-003	BR-03	Student attempts HOD Portal access	HOD URL	Enter HOD URL directly	Access is denied.	Critical
TC-RBAC-004	BR-03	Student attempts Super Admin access	Admin URL	Enter Admin URL	Access is denied.	Critical
TC-RBAC-005	BR-05	Faculty accesses assigned academic data	Assigned subject/section	Open assigned subject	Faculty can view/manage authorized academic data.	Critical
TC-RBAC-006	BR-05	Faculty accesses another unauthorized subject	Unassigned subject	Attempt to open/manage unassigned subject	Access is denied and unauthorized data is not exposed.	Critical
TC-RBAC-007	BR-06	HOD accesses own department	Authorized department	Open department data	HOD can access authorized departmental information.	High
TC-RBAC-008	BR-06	HOD accesses another department	Unauthorized department	Attempt to access another department	Access is denied.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-RBAC-009	BR-03	Super Admin accesses administrative functions	Admin account	Open admin portal	Authorized administrative functions are available.	Critical
TC-RBAC-010	BR-17	User attempts unauthorized modification	Record outside authorization scope	Modify record through UI/API	Modification is rejected and unauthorized data remains unchanged.	Critical

3. Student Portal Test Cases

The BRD defines student capabilities including dashboard, academic information, courses, resources, assignments, examinations, attendance, results, productivity tools, calendar, communication, online classes, notifications, AI support, study groups, exam-focused study, and profile/security.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STU-001	F01	Load student dashboard	Student account with academic data	Open Dashboard	Dashboard loads personalized academic information for the student.	High
TC-STU-002	F01	Verify dashboard student identity	Student profile	Open dashboard	Correct student name/profile information is displayed.	High
TC-STU-003	F02	View academic section	Program/semester/section	Open Academic Section	Student's authorized academic information is displayed.	High
TC-STU-004	F03	View courses/subjects	Subject list	Open Courses	Only subjects associated with student's academic context are displayed.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STU-005	F04	View study resources	PDF/document	Open Study Resources	Available authorized learning materials are displayed.	High
TC-STU-006	F04	Download learning resource	Valid resource	Select Download	Authorized file downloads successfully.	Medium
TC-STU-007	F05	View assignment list	Assignment records	Open Assignments	Relevant assignments and their status/deadlines are displayed.	High
TC-STU-008	F08	View examination schedule	Exam data	Open Exams	Correct authorized examination schedule is displayed.	High
TC-STU-009	F09	View attendance	Attendance data	Open Attendance	Student can view attendance applicable to their academic records.	High
TC-STU-010	F10	View published result	Published result	Open Results	Published result is displayed correctly.	Critical
TC-STU-011	F10 / BR-09	View unpublished result	Unpublished result	Open Results	Result is not displayed to the student.	Critical
TC-STU-012	F12	Create personal task	Task title/date	Create task	Task is created and appears in student's task list.	Medium
TC-STU-013	F13	View academic calendar	Event/class/exam	Open Calendar	Relevant academic events are displayed according to student's scope.	Medium

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STU-014	F16	View notifications	Notification record	Open Notifications	Student can view relevant notifications.	High
TC-STU-015	F20	View profile	Student profile	Open Profile	Authorized profile information is displayed.	Medium

4. Faculty Portal Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-FAC-001	F21	Faculty dashboard loads	Faculty account	Open Dashboard	Faculty dashboard loads with authorized academic information.	High
TC-FAC-002	F22	View academic mapping	Subject/section mapping	Open Academic Mapping	Assigned subjects, sections and relevant academic mappings are displayed.	High
TC-FAC-003	F23	Upload learning material	Valid PDF/document	1. Open Materials. 2. Select subject. 3. Upload file.	File is uploaded and associated with the correct academic context.	High
TC-FAC-004	F23	Upload unsupported material	Unsupported file	Upload file	Upload is rejected with validation message.	High
TC-FAC-005	F24	Create assignment	Title, description, deadline	Create assignment	Assignment is created with correct subject and deadline.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-FAC-006	F24	Publish assignment	Draft assignment	Select Publish	Assignment status changes to published and becomes available to authorized students.	Critical
TC-FAC-007	F25	View student submission	Student submission	Open submissions	Faculty can view submissions belonging to authorized students.	High
TC-FAC-008	F25	Evaluate assignment	Marks + feedback	Enter marks and feedback	Evaluation is saved against the correct submission.	Critical
TC-FAC-009	F26	Record attendance	Student attendance	Mark students present/absent	Attendance is saved against correct class/date/student records.	Critical
TC-FAC-010	F27	Enter examination marks	Valid marks	Enter marks and save	Marks are stored against correct student and assessment.	Critical
TC-FAC-011	F27	Enter invalid marks	Negative/over-limit marks	Enter invalid marks	System rejects invalid marks and displays validation message.	High
TC-FAC-012	F28	Schedule online meeting	Meeting details	Create online session	Meeting is created and associated with correct academic context.	High
TC-FAC-013	F29	Send faculty communication	Message	Compose and send message	Message is delivered to authorized recipients.	Medium

5. HOD / Domain Admin Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-HOD-001	F30	HOD dashboard loads	HOD account	Open dashboard	Dashboard displays authorized departmental information.	High
TC-HOD-002	F31	Allocate faculty to subject	Faculty + subject	Select faculty and assign subject	Faculty allocation is saved successfully.	Critical
TC-HOD-003	F31	Allocate unauthorized faculty	Faculty outside scope	Attempt assignment	System prevents unauthorized allocation.	Critical
TC-HOD-004	F32	Monitor students	Department students	Open student monitoring	HOD can view students belonging to authorized department.	High
TC-HOD-005	F33	Monitor faculty	Department faculty	Open faculty monitoring	HOD can view authorized faculty information.	High
TC-HOD-006	F34	Create/update timetable	Class schedule	Add/update timetable entry	Timetable is saved and reflected for authorized users.	High
TC-HOD-007	F35	Approve academic request	Request record	Review and approve	Request status changes to approved and appropriate users are notified if configured.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-HOD-008	F36	Generate department report	Report parameters	Select report criteria	System generates report containing authorized departmental data.	Medium

6. Super Admin Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ADM-001	AD-01	Create student account	Student basic information	Open User Management → Create Student → enter details → Save	Student account is created with correct academic mapping.	Critical
TC-ADM-002	AD-01	Create faculty account	Faculty details	Create Faculty → enter details → Save	Faculty account is created successfully.	Critical
TC-ADM-003	AD-01	Deactivate user	Existing user	Open user → Deactivate	User is marked inactive and cannot authenticate afterward.	Critical
TC-ADM-004	AD-02	Create department	Department name	Add Department → Save	Department is created successfully.	High
TC-ADM-005	AD-02	Create program	B.Tech/BBA/MBA	Add Program → select department → Save	Program is created under correct department.	High
TC-ADM-006	AD-02	Create semester	Semester data	Add Semester → Save	Semester is created under correct program.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ADM-007	AD-02	Create section	Section data	Add Section → Save	Section is created under correct academic hierarchy.	High
TC-ADM-008	AD-02	Create subject	Subject details	Add Subject → Save	Subject is created and associated with correct academic structure.	High
TC-ADM-009	AD-03	Assign user role	Student/Faculty/HOD/Adm in role	Select user → Assign role	Correct role is assigned and corresponding permissions become effective.	Critical
TC-ADM-010	AD-06	View audit logs	Critical action	Open Audit Logs	Authorized administrator can view relevant audit entries.	High

7. Student Account & Credential Test Cases

The platform requirement is that users are provisioned by authorized administrators rather than through unrestricted public registration.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-USER-001	BR-02	Create student using valid information	Student name, college email, program, semester, section	Fill student form and submit	Student account is created successfully.	Critical
TC-USER-002	BR-02	Create student using duplicate email	Existing email	Enter duplicate email and submit	System rejects duplicate email and does not create duplicate account.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-USER-003	BR-02	Create student with missing required information	Missing required field	Submit form	Required-field validation is displayed and account is not created.	High
TC-USER-004	BR-02	Temporary password generation	Valid student data	Create account	System generates/provisions a temporary credential according to configured process.	Critical
TC-USER-005	BR-02	Credential email delivery	Student email	Create student	Student receives required account information through configured email mechanism.	High
TC-USER-006	BR-02	Credential SMS delivery	Student mobile number	Create student	Student receives required account information through configured SMS mechanism.	High
TC-USER-007	BR-02	First login with temporary password	Temporary credential	Login using temporary credential	User can authenticate according to configured first-login flow.	Critical
TC-USER-008	BR-02	Change temporary password	New password	Enter new password and save	Temporary credential is replaced according to password policy.	Critical
TC-USER-009	BR-01	Student attempts public registration	Registration URL	Attempt to create account independently	Unrestricted public account creation is not available.	Critical

Important: The exact password policy, SMS provider, email provider, and retry behavior should be recorded as open requirements if they are not defined in the BRD/SRS rather than invented.

8. Assignment Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ASG-001	F24	Faculty creates assignment	Assignment title, description, deadline	Enter assignment details → Save	Assignment is created successfully.	Critical
TC-ASG-002	F24	Create assignment without title	Blank title	Submit assignment	System displays required-field validation.	High
TC-ASG-003	F24	Publish assignment	Valid assignment	Click Publish	Assignment becomes visible to authorized students.	Critical
TC-ASG-004	F05	Student views published assignment	Assignment	Open Assignments	Assignment details and deadline are displayed.	High
TC-ASG-005	F05	Student submits assignment	Valid file/submission	Open assignment → Upload → Submit	Submission is saved successfully and status changes appropriately.	Critical
TC-ASG-006	F05	Student submits duplicate submission	Second submission	Submit again	System follows configured resubmission rule and prevents unintended duplicate records.	High
TC-ASG-007	BR-16	Student submits after deadline	Late submission	Attempt submission	System applies the configured late-submission rule and displays appropriate status/message.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ASG-008	F25	Faculty evaluates submission	Marks + feedback	Open submission → Enter evaluation → Save	Marks and feedback are stored correctly.	Critical

TC-ASG-009	F25	Student views faculty feedback	Feedback	Open evaluated assignment	Student can view authorized feedback and marks.	High
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9. Attendance Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ATT-001	F26	Faculty records attendance	Student attendance list	Open class → Mark attendance → Save	Attendance is stored against correct students/date/class.	Critical
TC-ATT-002	F09	Student views attendance	Student record	Open Attendance	Correct attendance information is displayed.	High
TC-ATT-003	BR-17	Student attempts to modify attendance	Attendance record	Attempt edit	Modification is rejected.	Critical
TC-ATT-004	BR-05	Faculty attempts unauthorized attendance modification	Unauthorized class	Attempt update	System denies modification.	Critical
TC-ATT-005	F26	Invalid attendance data	Invalid/missing attendance value	Submit invalid record	System validates and rejects invalid attendance data.	High